



BEYOND FORGETTING: Diagnosing and Harnessing Shared Reasoning in Continual RLVR

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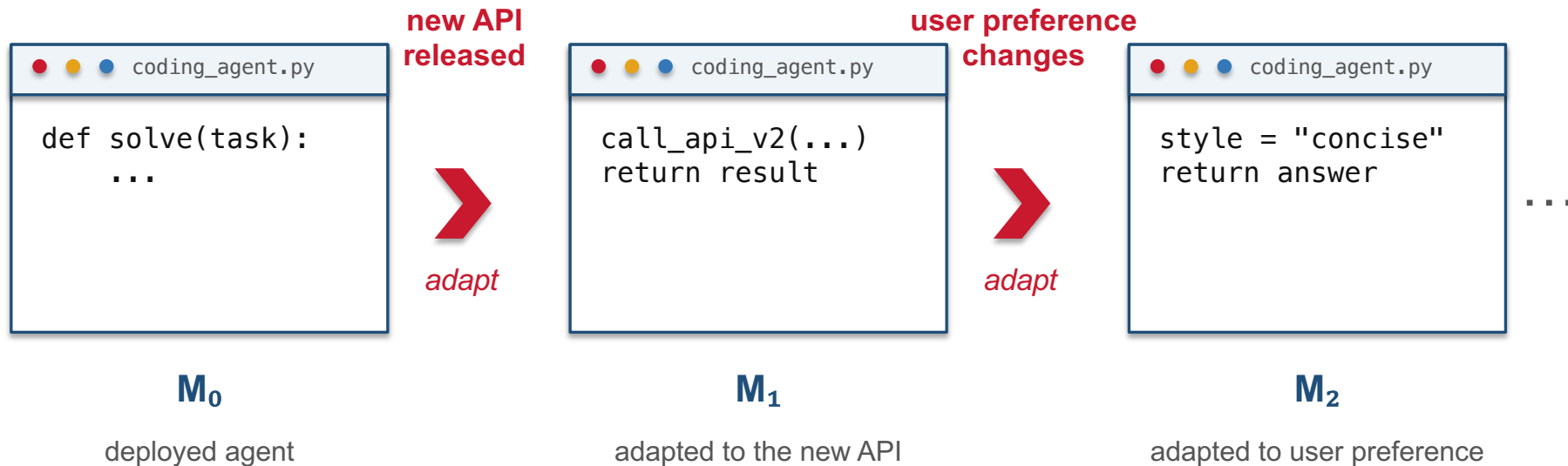
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<https://crg-rl.github.io/> · arXiv:2608.18574 · github.com/crg-rl/crg

Seq. RLVR forgets modestly yet reaches only **0.88× MTRL**. CPR reaches **1.03×**.

5 task sequences · 30 stages · text and visual reasoning

A deployed coding agent must keep adapting



New capability requirements keep arriving after deployment.

Joint retraining becomes costly as capabilities grow

Joint retraining after every task arrival



retrain the enlarged mixture at every step

Task-by-task update

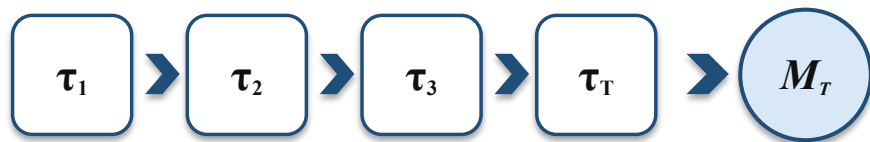


reuse the existing policy

Continual RLVR aims to match joint-training performance
through task-by-task updates.

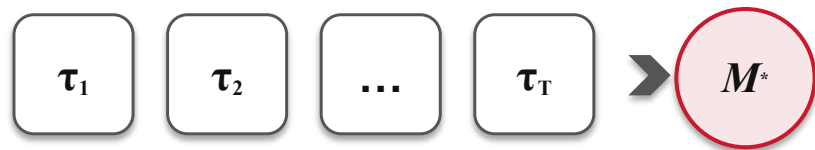
Can task-by-task RLVR match joint training?

Continual RLVR



tasks arrive one by one

Multitask RLVR (MTRL)

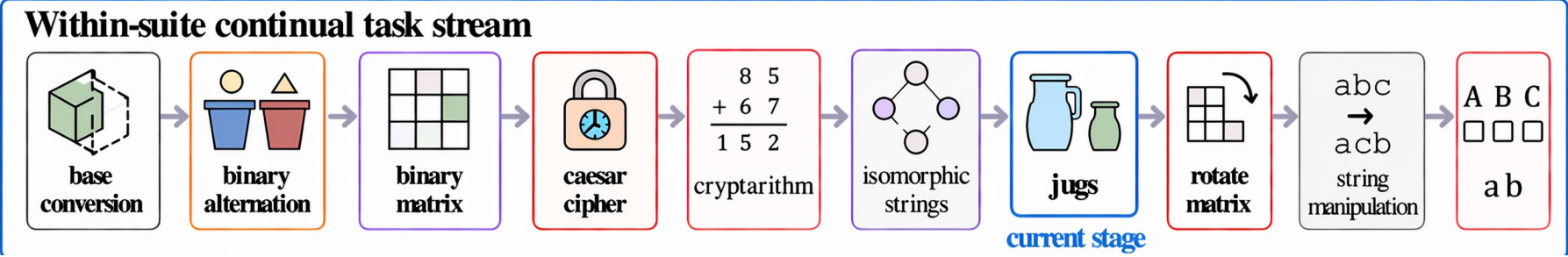


joint optimization

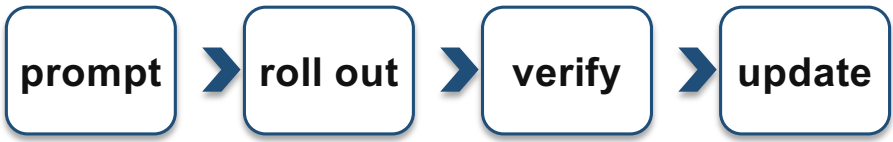
$$\text{CTM} = \frac{\text{FinalAvg}(M_T)}{\text{FinalAvg}(M^*)} \rightarrow 1$$

CTM compares the continual endpoint with its joint-training reference.

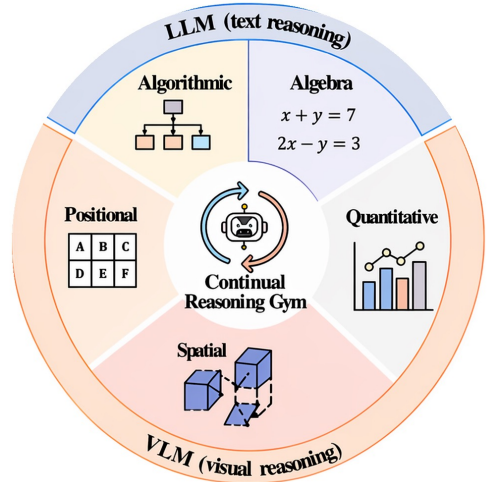
CRG instantiates five continual RLVR streams



At each stage

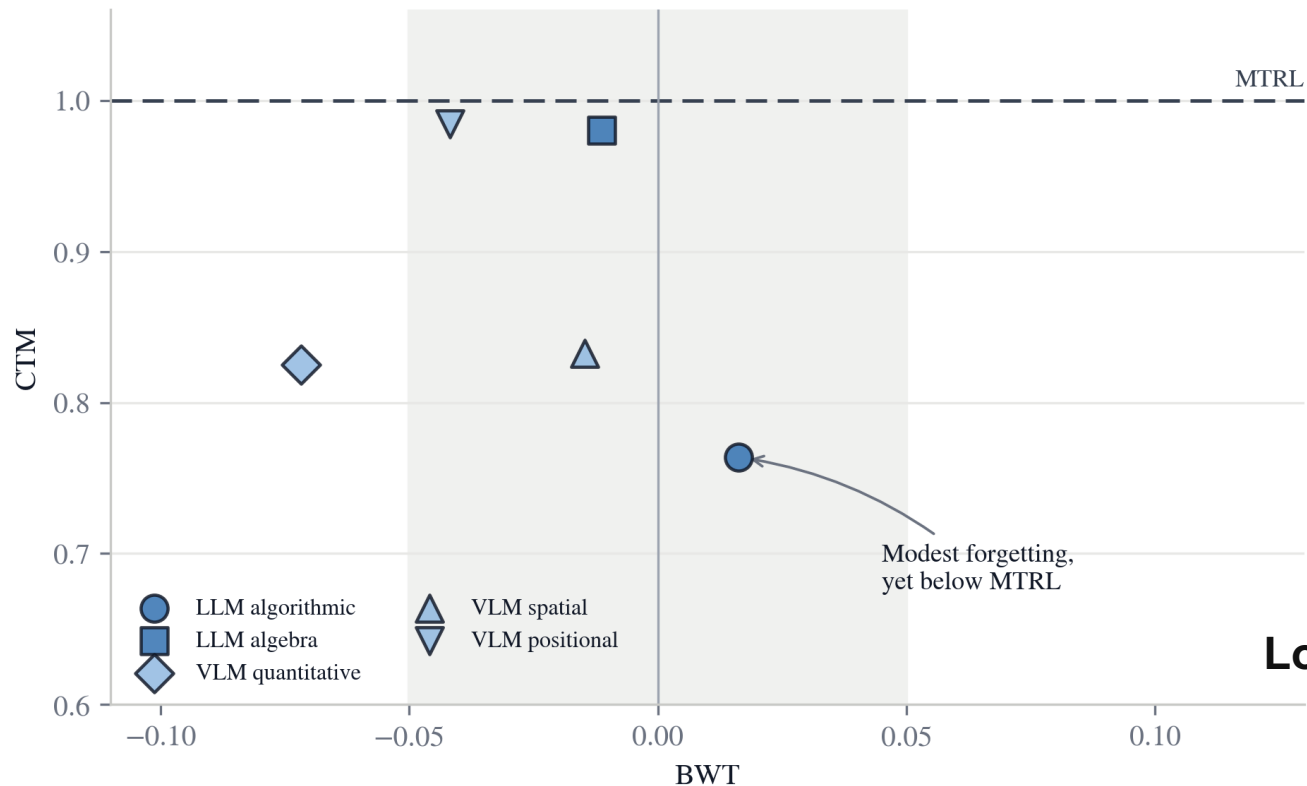


then move to the next task stage



30 stages across two text-reasoning and three visual-reasoning sequences.

Seq. RLVR forgets modestly yet reaches only 88% of MTRL



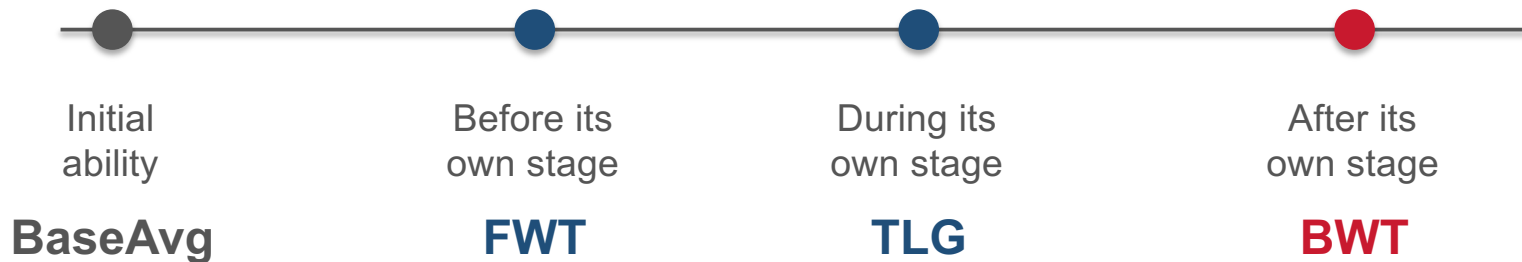
-2.47 pp
mean backward transfer

0.88
mean CTM

**Low forgetting does not imply
MTRL-level performance.**

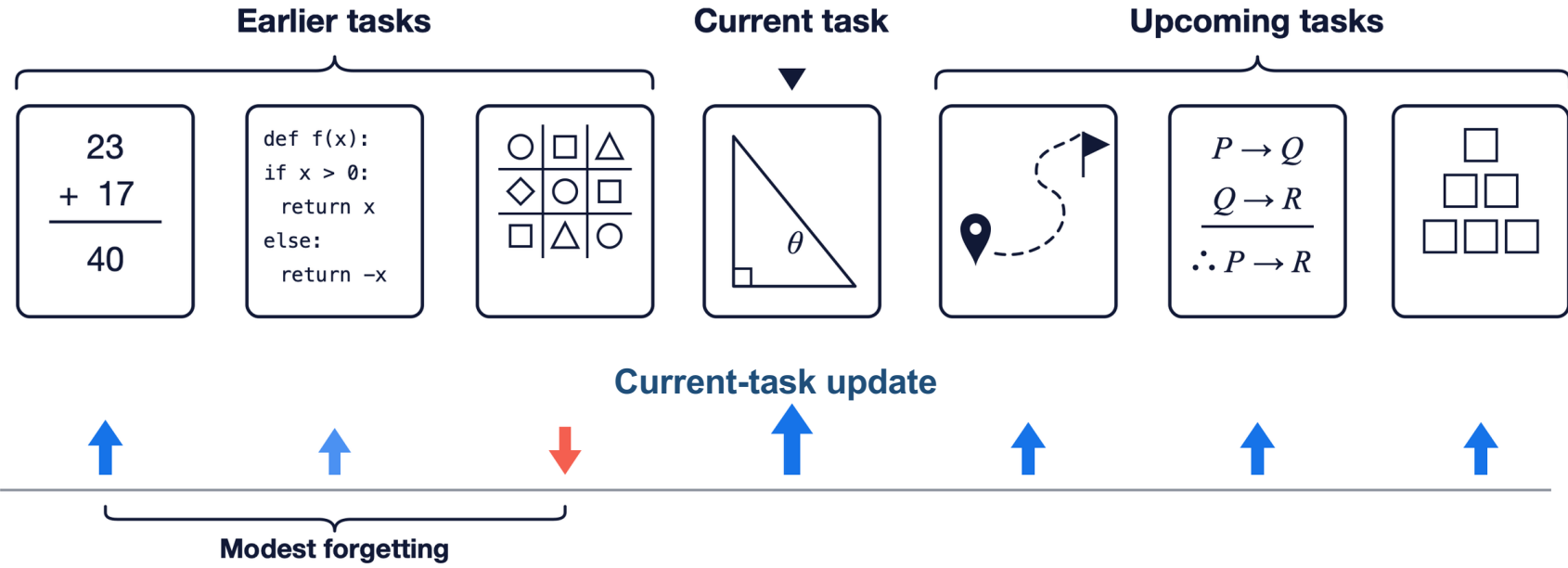
Forgetting explains only part of the MTRL gap

$$\mathbf{FinalAvg} = \mathbf{BaseAvg} + \frac{T-1}{T} \mathbf{FWT} + \mathbf{TLG} + \frac{T-1}{T} \mathbf{BWT}$$



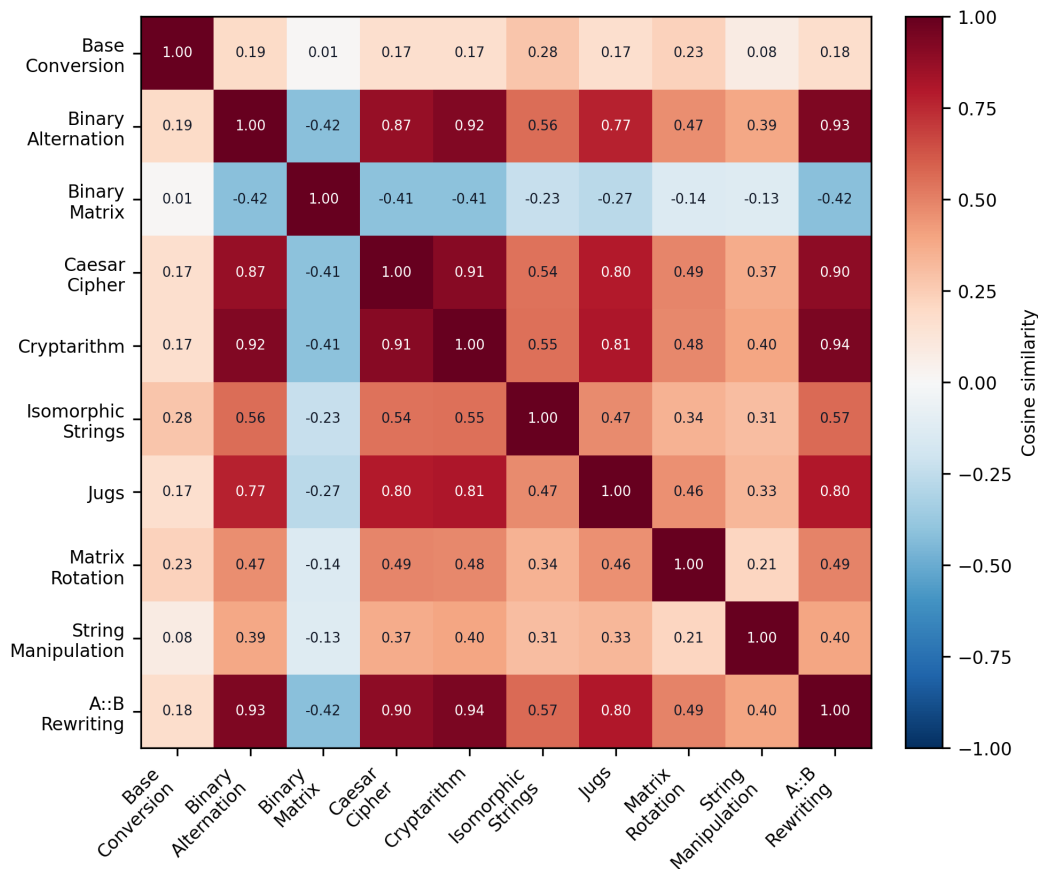
Even eliminating forgetting leaves gaps in **forward transfer** and **direct task learning**.

Shared reasoning explains modest forgetting



Shared reasoning: an update on one reasoning task benefits others on average. Later-task updates therefore limit net forgetting.

Task gradients are positively aligned on average



0.345

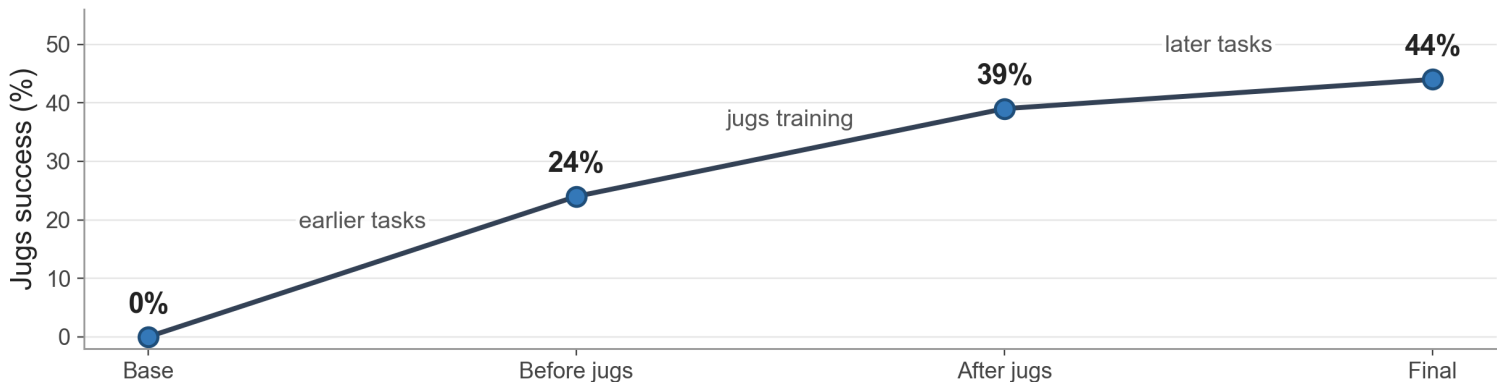
mean pairwise cosine

82.2%

positive task pairs

The positive mean is consistent with shared reasoning.

Shared reasoning also appears in model behavior



Base

*"Maybe fill B or C ... Wait ...
There we go! ... But let me—"*

**Second-guesses
a valid route**

No answer

Before jugs

*"Maybe use the 11L jug ...
that's not getting me to 3.
Let me try another approach ..."*

Finds a longer route

No answer

After jugs

*"I have a valid route ...
let me track each step ...
B has 3."*

Rechecks the route

Valid answer

Final

*"Fill B. Pour into A ... empty A ...
fill B again ... B has 3."*

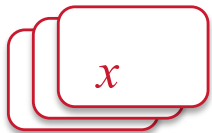
Direct reasoning

Valid answer

Earlier- and later-task training improve both success and reasoning.

CPR brings previous tasks into current-policy training

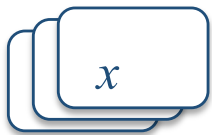
previous-task
prompts



**mixed prompt
batch**

replace, do not add

current-task
prompts



current policy

$$\pi_{\theta}(y \mid x)$$

fresh responses

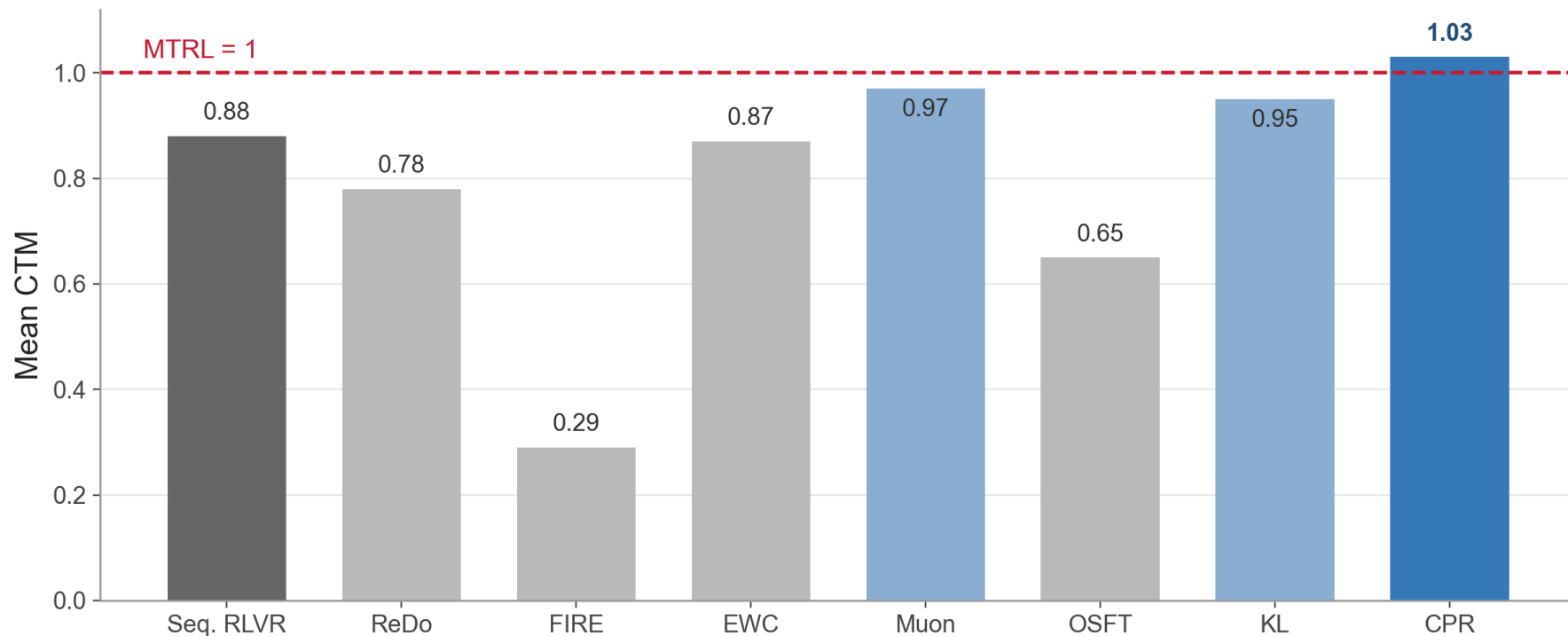
**task
verifier**

reward

**RLVR
update**

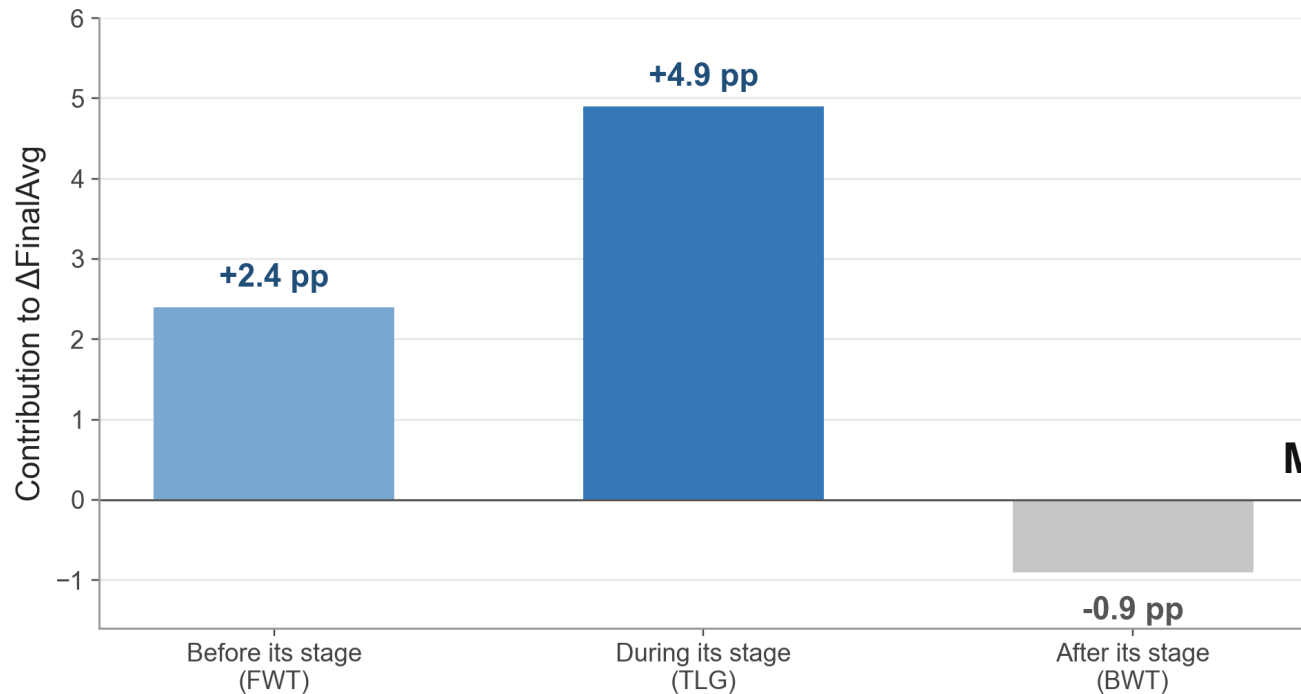
CPR changes the prompt distribution **without increasing prompt or rollout counts.**

Only CPR reaches MTRL-level performance on average



CPR reaches 1.03 mean CTM, compared with 0.88 for Seq. RLVR.

CPR improves learning on arriving and future tasks



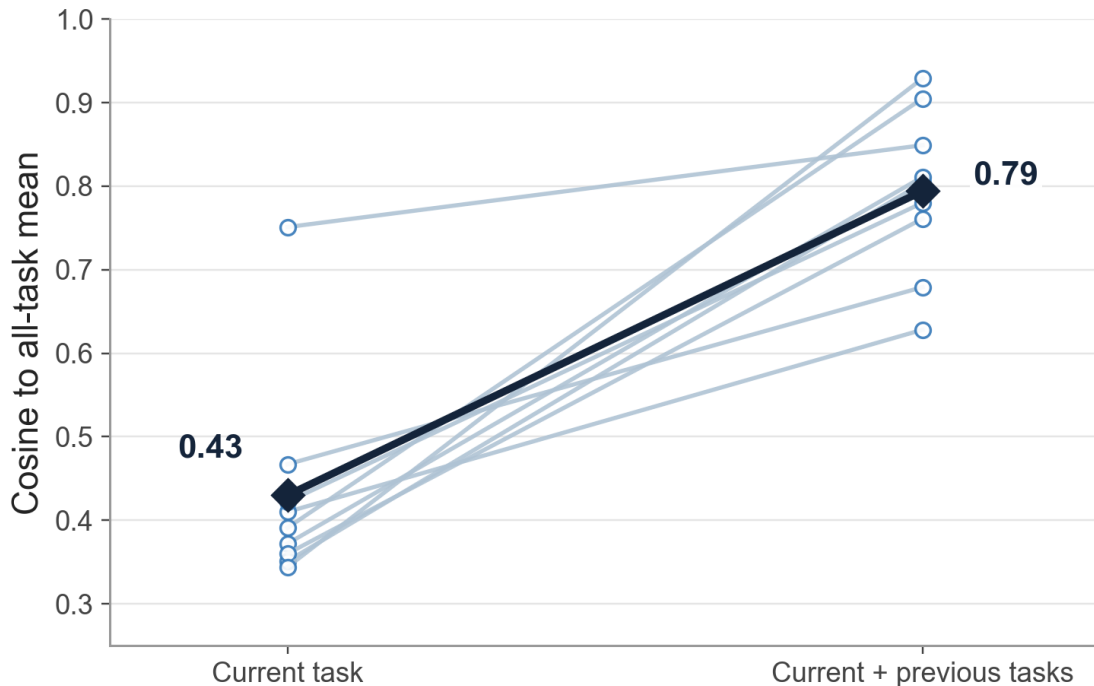
+6.4 pp

mean FinalAvg gain



**Most of the gain comes from
direct task learning
and forward transfer.**

Previous-task gradients improve alignment with the full task set



0.43 → 0.79

cosine to the all-task mean

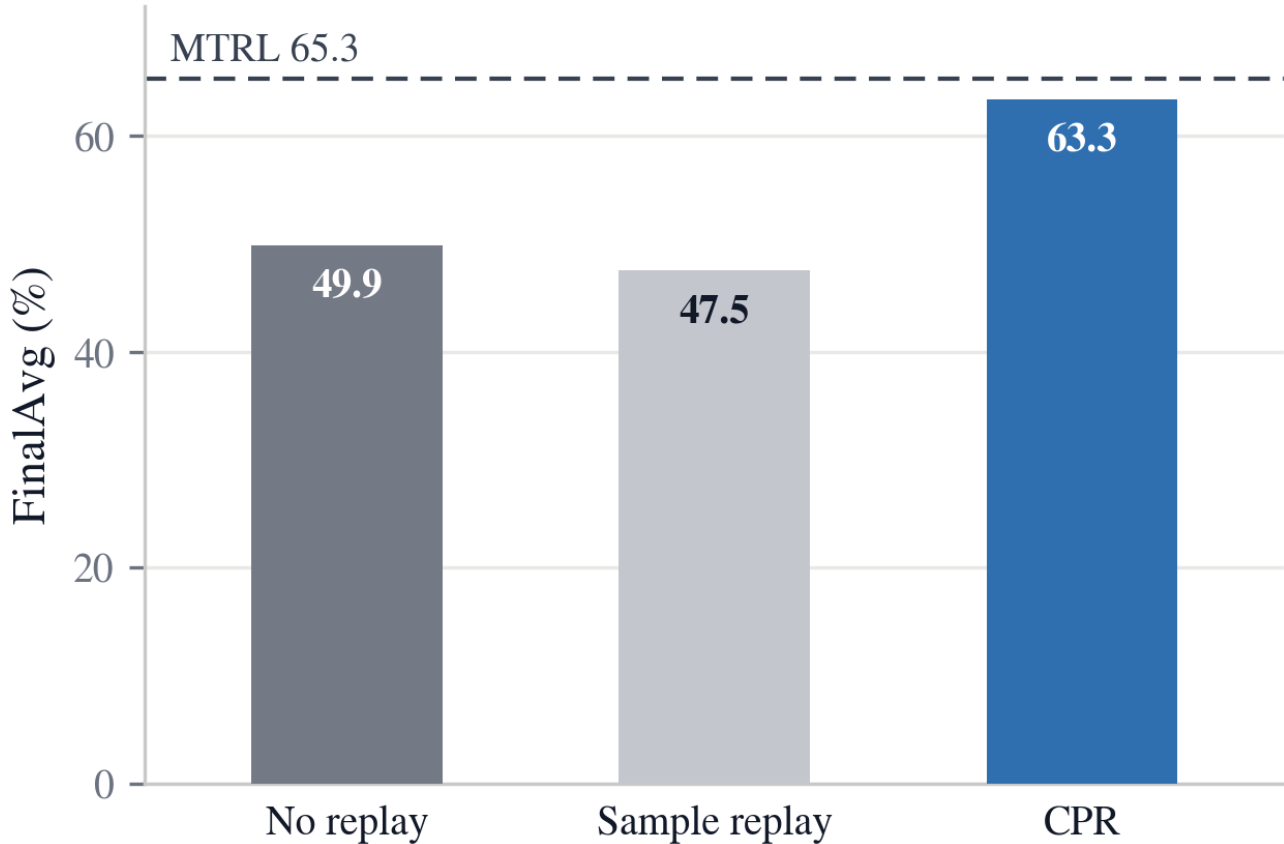
9 / 9

stage comparisons improve

The mixed direction is closer to the gradient of the full task set.



Current-policy regeneration is critical



63.3
CPR FinalAvg

47.5
sample replay

**Replay prompts.
Regenerate responses.**



Takeaways

- **Seq. RLVR** forgets modestly, yet reaches only 0.88 mean CTM.
- **Shared reasoning** explains modest forgetting and motivates CPR.
- **CPR** harnesses shared reasoning and reaches 1.03 mean CTM.

Project: <https://crg-rl.github.io/> **Code:** github.com/crg-rl/crg

Beyond Forgetting

Diagnosing and Harnessing Shared Reasoning in Continual RLVR

